

BANARAS HINDU UNIVERSITY

B.A. (Hons.) — Semester I Examination, 2022-23

Paper No. MTB-101: Calculus-I

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 055823

*Instructions: Answer **any five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Define a bounded sequence with a suitable example.
- (b) Explain convergent and divergent sequences with suitable examples.
- (c) Prove that if $\sum_{n=1}^{\infty} u_n$ converges, then $\lim_{n \rightarrow \infty} u_n = 0$.
- (d) State D'Alembert's ratio test for $\sum_{n=1}^{\infty} u_n$ of positive real numbers.
- (e) Find the asymptotes of $4x^2 + 9y^2 = x^2y^2$.
- (f) Find the point(s) of inflection of $f(x) = x^4 + 2x^3$.
- (g) Find the n th order derivative of $y = \sin(ax + b)$, $a \neq 0$, $b \neq 0$.
- (h) Determine the critical points of $f(x) = 6 + 12x + 3x^2 - 2x^3 - 4x^4$, $x \in \mathbb{R}$.
- (i) Elaborate the Sandwich theorem for sequences.
- (j) Show that $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ is convergent.
- (k) State Taylor's theorem with Lagrange's form of remainder.

Question 2.

- (a) Prove that every monotonically decreasing, bounded-below sequence converges. Illustrate with an example. [6]
- (b) Test convergence of $\sum_{n=1}^{\infty} \frac{n}{(1+n^4)^{1/2}} x^n$, $x > 0$. [6]

Question 3.

- (a) State and prove the Intermediate Value Theorem for continuous functions. [6]
- (b) Prove, using ε - δ definition: (i) $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$; (ii) $\lim_{x \rightarrow 4} \frac{\cos(\pi x + 5)}{x}$. [6]
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Question 4.

- (a) State and prove Rolle's theorem. Verify for $f(x) = x^2 - 5x + 6$, $x \in [2, 3]$. [6]
- (b) State and prove Cauchy's mean value theorem. Deduce Lagrange's MVT from it. [6]
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Question 5.

- (a) Examine continuity and differentiability of $f(x) = \begin{cases} x^2 \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$ at $x = 0$. [6]
- (b) Explain the geometrical interpretation of Lagrange's MVT. Verify for $f(x) = x(2 - x)$, $x \in [0, 1]$. [6]
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Question 6.

- (a) Apply Maclaurin's theorem to prove $\log \sec x = \frac{x^2}{2} + \frac{x^4}{12} + \dots$ [6]
- (b) State and prove Leibnitz's theorem for successive differentiation of $u(x) \cdot v(x)$. [6]
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Question 7.

- (a) If $y = a \cos(\log x) + b \sin(\log x)$, show $x^2 y_{n+2} + (2n + 1)xy_{n+1} + (n^2 + 1)y_n = 0$. [6]
- (b) Sketch the graphs of: (i) $y = e^x + e^{-x}$; (ii) $y = a(1 - \cos \theta)$, $a > 0$. [6]
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